COMPUTER ORGANIZATION AND ARCHITECTURE
ASSIGNMENT - I

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1. What is raspberry pi?

Ans: Raspberry pi which was made in United Kingdom (UK), is a small affordable single board computers which was made by the Raspberry Pi Foundation. People from all over the world use this small computer to build projects i.e hardware projects, for learning Programming skills, to automate home appliances and many more. Unlike the traditional computers, a Raspberry pi is the size of a credit card, but contains all the essential computing components in the single board. It mainly runs on Linux and has a set of input output pins for controlling the electronic systems.

2. AI Summit 10 key takeways.

Ans: The key take-ways of AI Summit 2026 are as follows:

i. **AI FOR ALL: " Sarvajan Hitaya, Sarvajan Sukhaya"** — Welfare for All, Happiness for All.
The main idea was that AI shouldn't be only benefiting big technological companies or any advanced country, it should also help farmers in detecting crop diseases, help students to learn any regional languages and help people with disabilities to communicate easily and many more.

ii. **Democratization of AI** : AI resources shouldn't be only controlled by countries or companies that can afford enormous computing structures. Thus called for expanding the access of AI to reduce the gap between countries and companies that have massive AI capacity and those who that don't including datasets, models and other AI resources in an affordable range.

iii. **HUMAN + AI : "AI will replace human"** - this is the main idea of max of the people but the summit talked about an era where the humans and AI will co-work, co-create and co-evolve.

iv. **M.A.N.A.V:**
M — Moral and Ethical Systems
A — Accountable Governance
N — National Sovereignty
A — Accessible and Inclusive
V — Valid and Legitimate
in simple language AI should be ethical, accountable, sovereign, accessible and legitimate.

v. **Multilingual AI:** India is a country with diverse culture and languages, but most of the AI technologies are mainly based only in English and other widely languages. Therefore, in the summit it was emphasized on inclusive and multilingual especially for the South.

vi. **AI for Economic Growth and Social Good :** When we use chatbots it simply looks like, Question --> AI --> Answer. But behind the screen it also uses severs, data centers, electricity and cooling. Thus large AI models require enormous electricity and water when serving huge numbers of users. Since,

India wants AI to grow massively, India's AI growth needs to happen alongside energy efficiency and sustainability.

vii. **India wants to become an AI innovation hub :** India doesn't only want to use AI products made by other countries but also it want to become a place where AI is built, tested and developed.

viii. **Building the Computing Infrastructure for AI :** India needs large amount of computing power, GPU, data centers, storage, electricity and cooling to develop modern AI.

ix. **The AI Economy Will Require Massive Investment :** In the summit it was made clear that AI is not just developing software models but building the AI economy itself requires enormous capital investment.The government reported investment commitments across the AI value chain exceeding **$250 billion** following the summit.

x. **International Cooperation Is Necessary :** No single country can solve the AI challenges alone. Thus in the summit it was strongly emphasized in international cooperation. The New Delhi Declaration was endorsed by 92 countries and international organizations, demonstrating broad support for cooperation around AI.

3. What is Yotta?

Ans: Yotta is a huge data service providing company in India which mainly has AI infrastructure and cloud computing which provide these services to companies, governments, researchers and also to AI developers to build and run some AI systems. It provides an AI focused cloud platform called the SHAKTI CLOUD, which provides access to high performance GPU computing for AI workloads.

4. What is NPTEL?

Ans: NPTEL stands for National Programme on Technology Enhanced Learning, which is an online learning platform created by famous and leading IIT and IISC to provide high quality educational courses which is free, mainly in engineering side.It also provides certificates at the end of the course after completing all the assignments and final exam of it.

5. How to decide a programming language to design an application and what are the parameters?

Ans: A programming language shouldn't only be choosed only by its popularity, but there are some parameters for choosing a programming language. These are as follows:

- **The type of Application:**

First identify what to build, for e.g

- Web application: JavaScript, Python.
- Desktop application: C++, C, Python, Java
- Game application: C++,C.

- **Performance requirements:**

- High performance: C, C++
- Rapid development where performance is less critical: Python, java

- **Libraries and Frameworks:**

- Python: AI/ML, data science, automation
- JavaScript/TypeScript: web development
- Java: enterprise applications
- C++: games, systems, high-performance applications
- C#: Windows/.NET applications and games

6. Ram size, Word size, meaning if system is of 64 bit, speed and os of our laptop.

Ans:

- Ram size (the computer's temporary working memory) : 16 GB
- Word size (the number of bits a CPU can process as a unit of data) : 64 - bit = 8 bytes,

which means that the computer's CPU and operating system are designed for working with 64 bit data value.

- Speed of a computer (how quickly the computer can perform operations and complete tasks): 2.10 GHz
- Operating system : 64 bit Windows.

7. Which CPU organization is my laptop?
Ans: Intel Core i5.

8. Why is GPU faster than CPU?
Ans: CPU is the general purpose brain of the computer, handles different types of task and controls overall operation in a system, whereas GPU is designed to perform multiple calculations simultaneously involving graphics and large amount of data.

9. CPU vs TPU vs GPU.
Ans:

	CPU	**TPU**	**GPU**
Full form	Central Processing Unit	Tensor Processing Unit	Graphics Processing Unit
Main purpose	Used for general tasks, i.e can do many different types of tasks	Used for AI specialized processing, i.e specially designed for AI calculations.	Used for parallel processing, i.e can do many similar tasks at the same time.
Best for	Everyday computing	Gaming, Graphics	Deep learning, AI
AI speed	Good	Very good	Excellent for supported AI
Developed by	Many companies	Google	NVIDIA, AMD, Intel

10. Who won Turing award last year?
Ans: In **2025**, the Turing Award which is one of the highest honors in computer science, often called the **"Nobel Price of Computing"** , was awarded to **Charles H. Bennett and Gilles Brassard** for their foundational contributions to quantum information science, particularly quantum cryptography. The award currently carries $1 million as a price.

11. What is Stampeding herd problem?
Ans: In technology, the Stampeding problem also known as the Thundering Herd Problem, happens when many users requests for accessing the same resource at the same time which is causing an overload to the system, thus making the system slow or unavailable. For e.g, 10000 customers trying to buy a popular product at the same time. The server has to process these requests at the same time, which can make the server overloaded. Thus, the website may become slow or may even crash.

12. Who is the founder of Intel and AMD?
Ans: Intel was founded in **July 18, 1968** by **Robert Noyce and Gorden Moore**. The head quarters is in **Santa Clara, California, USA**.
AMD (Advanced Micro Devices, Inc) was founded in **May 1, 1969** by **Jerry Sanders,** along with seven co-founders. The headquarters is in **Santa Clara, California, USA**.